Java Programming

CPT111 – Lab 6

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CPT111 Java Programming

Lab 6

Introduction to Objects

Exercise Week #6

- Complete the class Clock that represents time on a 24-hour clock, such as 00:00, 15:30, or 23:59
  - Time is measured in hours (00 – 23) and minutes (00 – 59)
  - Times are ordered from 00:00 (earliest) to 23:59 (latest)
  - A skeleton file Clock.java is given

Exercise #6.1 Clock Constructor 1

- Complete the first constructor of the class Clock
- It takes two arguments: h and m
  - and creates a new clock object whose initial time is h hours and m minutes

- Test cases:

```
Clock clock1 = new Clock(1, 0);

System.out.println(clock1);                    → 01:00
```

Exercise #6.2 Clock Constructor 2

- Complete the second constructor of the class Clock
- It takes one string argument: s
  - s is composed of two digits, followed by a colon, followed by two digits, so the format is HH:MM such as 02:30
  - it creates a new clock object whose initial time is HH hours and MM minutes

- Test cases:

```
Clock clock2 = new Clock("02:30");

System.out.println(clock2);                    → 02:30
```

Exercise #6.3 Clock toString

- Complete the method toString of the class Clock
- It returns a string representation of this clock, using the format HH:MM
  - that is, the format is the hours (2 digits), followed by a colon, followed by the minutes (2 digits), for example, 00:00 and 23:59

- Test cases:

```
Clock clock1 = new Clock(1, 0);
Clock clock2 = new Clock("02:30");

System.out.println(clock1);          → 01:00
System.out.println(clock2);          → 02:30
```

Exercise #6.4 Clock isEarlierThan

- Complete the method isEarlierThan of the class Clock
- It returns true if and only if the time on *this* clock (the current referenced object) is earlier than the time on *that* clock (in the argument)

- Test cases:

```
Clock clock1 = new Clock(1, 0);
Clock clock2 = new Clock("02:30");

System.out.println(clock1.isEarlierThan(clock2));      → true
```

Exercise #6.5 Clock tick

- Complete the method tick of the class Clock
- It adds 1 minute to the time on this clock
  - for example, one minute after 01:00 is 01:01; one minute after 23:59 is 00:00

- Test cases:

```
Clock clock1 = new Clock(1, 0);
System.out.println(clock1);                    → 01:00

clock1.tick();
System.out.println(clock1);                    → 01:01
```

Exercise #6.6 Clock tock

- Complete the method tock of the class Clock
- It adds delta minute(s) to the time on this clock, where delta is a positive integer
  - for example, 100 minutes after 02:30 is 04:10
- Note that must **not** use the method in CW1 #7.5 tick().

- Test cases:

```
Clock clock2 = new Clock("02:30");
System.out.println(clock2);                    → 02:30

clock2.tock(100);
System.out.println(clock2);                    → 04:10
```

Lab 6 Challenge – Temperature Class

- Complete the class `Temperature` that stores a temperature internally in **Celsius** and provides methods to read or set it in either **Celsius** or **Fahrenheit**.
- A skeleton file `Temperature.java` is given

Lab 6 Challenge – Requirements

1. One private instance variable: `private double celsius`
2. A no-argument constructor that sets the temperature to 0.0 °C.
3. A one-argument constructor `Temperature(double celsius)` that sets the initial value in Celsius.
4. Getters:
   - `public double getCelsius()` — returns the value in Celsius.
   - `public double getFahrenheit()` — returns the equivalent value in Fahrenheit using this formula: F = C * 9/5 + 32.
5. Methods/behaviour:
   - `public void increase(double deltaC)` — adds `deltaC` degrees to the current Celsius value (can be negative).
   - `public boolean isBelowFreezing()` — returns `true` if the temperature is **strictly less than** 0.0 °C, `false` if not.
   - `public String toString()` — returns a string like `"0C (32F)"` (round to the nearest integer).

Lab 6 Challenge – Test Cases

```
// Example 1: default constructor + getters + toString
Temperature t0 = new Temperature();          // 0.0°C
System.out.println(t0.getCelsius());         // -> 0.0
System.out.println(t0.getFahrenheit());      // -> 32.0
System.out.println(t0);                      // -> 0C (32F)


// Example 2: increase and freezing check
Temperature t1 = new Temperature(-1.5);
System.out.println(t1.isBelowFreezing());    // -> true
t1.increase(3.0);                            // now 1.5°C
System.out.println(t1.isBelowFreezing());    // -> false
System.out.println(t1.getFahrenheit());      // -> 34.7 (approximately)
```

Thank you for your hard work!

- If you finish early, feel free to use the remaining time to work on other online exercises
  - Use this opportunity to practice your new skills in testing your code!
- You may leave the lab once you have completed the Lab 6 Challenge and it has been ASSESSED by a TA!
  - Please note that the **Lab 6 Challenge** is **graded** and contributes to your CW2 marks